Tensors of Minimal Border Rank

Arpan Pal

Texas A&M University, College Station, TX

Aug 17, 2021

Simultaneously Diagonalizable Matrices

Let $M_1, \cdots, M_m$ be m simultaneously diagonalizable $m \times m$ matrices.

Classical Problem

Characterizing the closure of the space of a finite number of simultaneously diagonalizable matrices.

Known Results

- End Closed Condition: Known since 1960.[Gerstenhaber]
- Flag Condition: Appeared in 1997.[Burgisser, Clausen, Shokrollahi]

Definitions

Let $T \in A \otimes B \otimes C$.
T can be seen as a linear map $T_A : A^* \to B \otimes C$. Similarly T_B and T_C.

Definition

The *rank* of a tensor T, denoted $\mathbf{R}(T)$, is the smallest r such that $T = \sum_{i=1}^{r} a_i \otimes b_i \otimes c_i$.

Definition

The *border rank* of a tensor T, denoted $\underline{\mathbf{R}}(T)$, is the smallest r such that $T = \lim_{\epsilon \to 0} T_\epsilon$ where $\mathbf{R}(T_\epsilon) = r$.

Definition

The tensor T is called *concise* if T_A, T_B and T_C are of full rank.

Questions

Assume $\dim(A) = \dim(B) = \dim(C) = m$ and $T \in A \otimes B \otimes C$.

Question 1

Find equations for the set of border rank at most m tensors.

Question 1 is answered up to dimension m=4.[Friedland]

Assuming T is concise, minimal possible border rank for T is m.

Question 2

Find equations for the set of concise minimal border rank tensors.

Under a natural genericity condition this question is same as characterizing the closure of simultaneously diagonalizable matrices.

Questions

$B \otimes C$ is identified with linear maps from B^* to C, Hom(B^*, C).

Definition

T is called 1_A-*generic*(similarly 1_B and 1_C) if there exist $x \in A^*$ such that $T_A(x)$ is invertible. T is called 1_*-*generic* if it is 1_A, 1_B or 1_C-generic.

If T is 1_*-generic then we reduce to the previous classical problem of classifying the closure of simultaneously diagonalizable matrices.

Question 3

Find equations for the set of concise, 1_*-generic, minimal border rank tensors.

Question 3 is answered for $m = 5$.[Landsberg, Michalek]

Motivation

- Question 1 is finding equations for the secant variety, $\sigma_m(\mathbb{CP}^{m-1} \times \mathbb{CP}^{m-1} \times \mathbb{CP}^{m-1})$.
- Complexity Theory: Latest bound on the exponent of matrix multiplication is achieved through Coppersmith-Winograd tensor. Which is a concise minimal border rank tensor.
- Classical Linear Algebra: Closure of simultaneously diagonalizable matrices.
- Algebraic Geometry: Hilbert schemes and Quot Schemes.

Strassen's Equations

Simultaneously diagonalizable $\implies$ Commuting.

Theorem ([Strassen])

Let X_1, X_2 and Y be in $T_A(A^*)$. If T is of minimal border rank then $X_1 adj(Y) X_2 - X_2 adj(Y) X_1 = 0$.

Assuming T 1_*-generic, we can take Y to be of full rank and in particular identity matrix. Then Strassen's Equations precisely reduces to commuting criterion of X_1 and X_2.

These are necessary conditions a tensor must satisfy in order to be of minimal border rank.

Koszul Flattenings

Let $T \in A \otimes B \otimes C$ and $\dim(A) = \dim(b) = \dim(C) = m$.

$$T^{\wedge p} : B^* \otimes \bigwedge^p A \xrightarrow{T_B \otimes Id} A \otimes \bigwedge^p A \otimes C \xrightarrow{\pi \otimes Id} \bigwedge^{p+1} A \otimes C$$

Theorem ([Landsberg, Ottaviani], Thm 2.1)

$\underline{R}(T) \geq \frac{rank(T^{\wedge p})}{\binom{m-1}{p}}$

Remark: $p = 1$ Koszul flattening equations are stronger than Strassen's equations.

Towards m=5 and 6, 1_A-generic case

In this case T is 1_A-generic and concise.

- Already solved, Question 3 for m=5.[Landsberg, Michalek]
- Known answer: Strassen's equations together with end closed condition.

Theorem (J. Jelisiejew, K. Sivic)

The closure of the space of 4-tuple of 5×5 commuting matrices is not irreducible and has exactly two components.
The principal component is the closure of simultaneously diagonalizable matrices and one other bad component.

Upshot: This extends to $m = 6$ and the same remains true!

Towards m=5, 1-degenerate

Theorem ([Friedland], Thm 3.1)

Let $T \in A \otimes B \otimes C$ be 1_A-degenerate and rank of elements of $T_A(A^*)$ are bounded by $m-1$ but not by $m-2$. Then there exist bases of A, B, C such that, letting $X_1, \cdots, X_m$ be a basis of $T_A(A^*)$ as a space of matrices,

1. $X_1 = \begin{pmatrix} Id_{m-1} & 0 \\ 0 & 0 \end{pmatrix}$
2. $X_m = \begin{pmatrix} \mathbf{x}_m & \omega \\ \alpha & 0 \end{pmatrix}$
3. *For all* $2 \leq s \leq m-1$, $X_s = \begin{pmatrix} \mathbf{x}_s & 0 \\ 0 & 0 \end{pmatrix}$.

Here $\omega \in \mathbb{C}^{m-1}$, $\alpha \in \mathbb{C}^{(m-1)*}$, $\mathbf{x}_m, \mathbf{x}_s \in Mat_{(m-1)\times(m-1)}$.
Moreover, $\alpha \mathbf{x}_m^j \omega = 0$ *for all* j, *and letting* $U_R = \langle \mathbf{x}_m^j \omega | j \in \mathbb{Z}_{\geq 0} \rangle \subset \mathbb{C}^{m-1}$ *and* $U_L = \langle \alpha \mathbf{x}_m^j | j \in \mathbb{Z}_{\geq 0} \rangle \subset \mathbb{C}^{(m-1)*}$. *Then* $\mathbf{x}_s U_R = 0 = U_L \mathbf{x}_s$ *for* $2 \leq s \leq m-1$.

Consequences:(Work in progress)

- Insisting $\mathbf{x}_5$ has distinct eigenvalues, gives three cases, almost done investigating them.
- Only a few cases arise depending on Jordan decomposition of $\mathbf{x}_5$.

Expectation: $p = 1$ Koszul flattening equations will be sufficient.

This is joint work(in progress) with Prof. JM Landsberg and Prof. Joachim Jelisiejew.

Thank you

Thank you!

I specially thank National Science Foundation(NSF) for supporting me for this conference.

References

Murray Gerstenhaber
On Dominance and Varieties of Commuting Matrices
Annals of Mathematics, Mar. 1961, Second Series, Vol. 73 No. 2, pp 324-348.

JM Landsberg, Mateusz Michalek
Abelian Tensors
Journal of Pure and Applied Mathematics, Vol. 108, Issue 3, September 2017, pp 333-371.

V. Strassen
Rank and Optimal Computation of Generic Tensors
Linear Algebra and its Applications, 52/53(1983), 645-685.

Joseph M. Landsberg, Giorgio Ottaviani
New Lower Bounds for the Border Rank of Matrix Multiplication
Theory of Computing, Vol. 11(11),2015,pp. 285-298.

Shmuel Friedland,
On tensors of border rank l in $\mathbb{C}^{m\times n\times l}$
Linear Algebra and its Applications, Volume 438, issue 2, pp 713-737

Peter Burgisser, Michael Clausen, and M. Amin Shokrollahi
Algebraic complexity theory